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Influence of interfering co-appearing container peaks on the accuracy of direct quantitative Raman measurement of a sample in a plastic container

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The axially perpendicular offset (APO) scheme was previously demonstrated as a versatile scheme able to minimize or eliminate the glass background in the direct and non-sampling Raman measurement of an ethanol sample housed in a glass bottle. Alternatively, when directly analyzing a sample housed in a plastic container, another typical container yielding strong Raman peaks itself, the Raman peaks of both the container and the housed sample are unavoidably present together in a collected spectrum. Therefore, a crucial issue to investigate under this situation is how the magnitude of the co-appearing container peaks influences the accuracy for quantitative analysis of the housed sample. For the evaluation, a non-sampling Raman spectroscopic measurement of the urea concentration in a urea gel housed in a circular polypropylene (PP) container was attempted by employing two axially perpendicular offset (APO) schemes with detection windows of different sizes (25.4 and 10.0 mm, referred to as the wide-window APO (WW-APO) and narrow-window APO (NW-APO), respectively), and transmission and back-scattering schemes incorporating a 25.4 mm detection window. The intensity ratios between the container and urea peaks in the collected spectra were different depending on the adopted measurement scheme. The intensity ratio was greatest (smallest container peak) in the NW-APO measurement due to the narrowed detection window, making the generated container Raman photons at the side-wall less detectable to the bottom-positioned detector. A spectral acquisition scheme allowing the maximal suppression of the container peaks, while still maintaining the sample features, was a key requirement to secure an accurate measurement of the sample concentration. In addition, a Monte Carlo simulation was used to visualize the distributions of the container and urea photons inside the sample-housed container.

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Introduction

As another measurement type enabling spatially offset Raman spectroscopy (SORS),[1,2] an axially perpendicular offset (APO) scheme incorporating an axially perpendicular geometric arrangement of the laser illumination and Raman photon detection[3,4] was demonstrated by this research group. The APO scheme was versatile in reducing or eliminating the glass background in the direct and non-sampling Raman measurement of an ethanol sample housed in a glass bottle. This was achieved by avoiding the detection of the glass photons with the positioning of the detector beneath the bottle at a proper offset distance.[3] The successful elimination of the glass background was facile because glass is a weak Raman scatterer, and its broad peak was rather localized to a certain spectral range.

In addition to glass containers, plastic containers are extensively used for sample housing. When a sample housed in a plastic container is directly analyzed using Raman spectroscopy, the complete suppression of the plastic container information in a collected Raman spectrum is expected to be difficult. The plastic peaks are usually intense and spread over a whole spectral range.[5–8] Moreover, the generated container Raman photons are expected to broadly spread in space due to the substantial isotropic diffusion of the photons inside a plastic medium by scattering.[8,9] Dissimilar to the measurement of a sample in a glass container in the previous report,

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the unavoidable plastic container peaks complicate the identity of the housed sample peaks and adversely affect the accuracy of the concentration determination.[10–12] Under this circumstance, a crucial issue to investigate is how the magnitude of the co-appearing container peaks influences the accuracy of the Raman measurement of a sample in a plastic container. This kind of systematic investigation has not been reported so far.

To address the issue, a non-sampling Raman spectroscopic measurement of the urea concentration (7.5–12.5%) in a urea gel housed in a circular polypropylene (PP) container was attempted by employing two axially perpendicular offset (APO) schemes with detection windows of different sizes (25.4 and 10.0 mm, referred to as a wide-window APO (WW-APO) and narrow-window APO (NW-APO), respectively), and the transmission and back-scattering schemes incorporating a 25.4 mm detection window. Fig. 1 shows a picture of the PP container and urea gel in the container, and also includes three-dimensional (3D) graphical descriptions of the four adopted Raman measurement schemes. Reduction of the container peaks would be more effective in the APO and transmission measurements since the detector is positioned farther from the point of laser illumination.[12–15] The degrees of the container peak reduction by these measurements are expected to differ.

Initially, the Raman spectral features of the urea gel housed in a PP container acquired with the four schemes were comparatively analyzed. This included analyzing the intensity ratio between the container and urea peaks, and the magnitude of the urea peak variation according to the change of the urea concentration. With the collected sample spectra in each case, the urea concentrations were determined using partial least squares (PLS), and the resulting accuracies were compared in

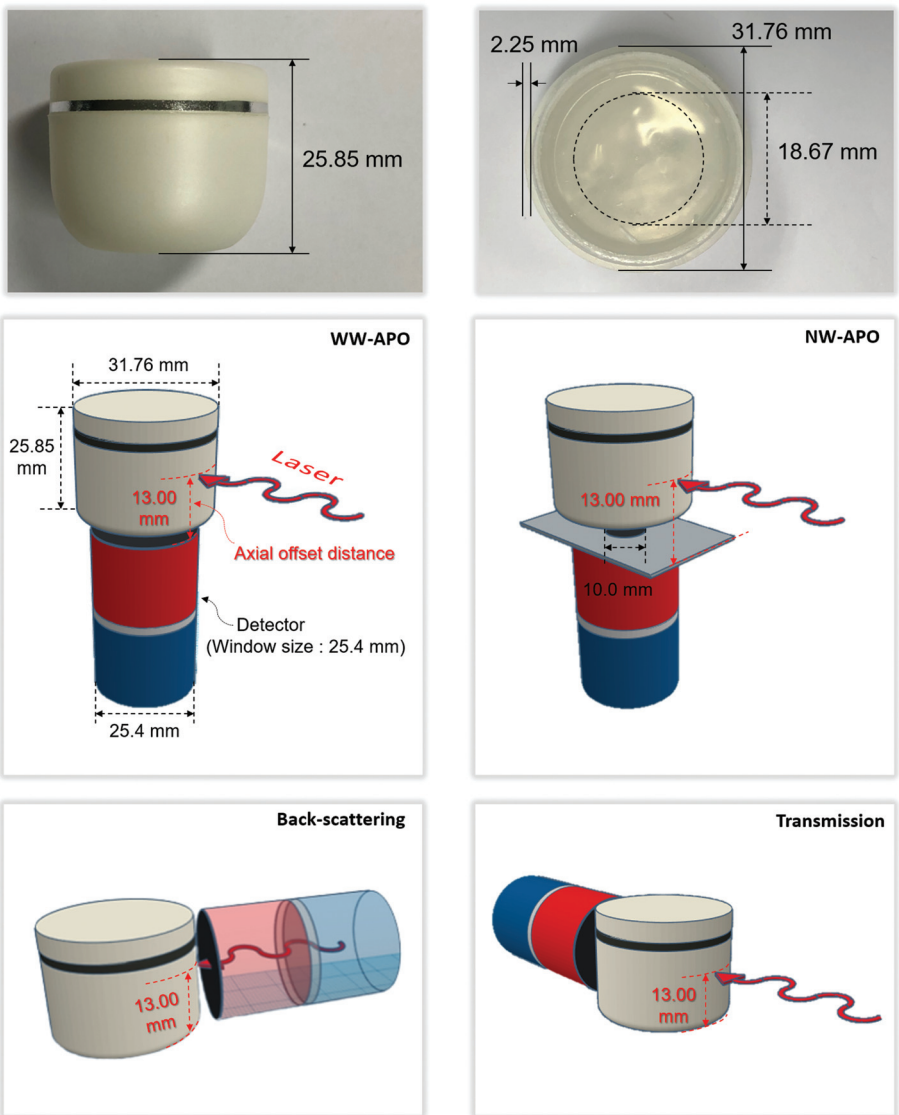

Fig. 1 Side view picture of the PP container (top left) and top view picture of the urea gel-filled container (top right). Three-dimensional graphical descriptions of the WW-APO, NW-APO, back-scattering, and transmission measurement schemes are also shown.

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relation to the relative magnitude of the container/sample peaks in the corresponding spectra. In addition, a Monte Carlo simulation was used to visualize the distributions of the container and urea photons inside the sample-housed container.

Experimental section

Preparation of urea gels

The urea gels used in this study were composed of carbopol 934 (0.4%), paraben concentrate (1.0%), propylene glycol (5%), NaOH (0.03%), urea, and purified water. The urea concentration varied from 7.5% to 12.5% with an increment of 0.5%. Carbopol 934 was initially dispersed in a mixture of propylene glycol, paraben concentrate, and purified water with constant stirring. Next, 2.5 M NaOH solution was added to the mixture with gentle stirring until it produced a homogeneous gel. Then, urea-dissolved purified water was added to the prepared gel with stirring to prepare a final urea gel sample. Cosmetic grade urea was used, and carbopol 934, propylene glycol, and paraben concentrate were purchased from BASF-Bangkok, Thailand. All reagents were of analytical grade, and distilled water was used throughout the sample preparation. Each final transparent urea gel sample was transferred into a PP plastic container (Fig. 1).

Acquisition of Raman spectra of the urea gel housed in a plastic container

Initially, the back-scattering Raman spectra of the urea gels housed in plastic containers were collected using a spectrometer (Raman Rxn1 unit, Kaiser Optical Systems, Ann Arbor, MI) equipped with a PhAT probe, providing a large illumination area (28.3 mm^2, diameter: 6 mm, wavelength: 785 nm) and a wide photon-collection area (506.7 mm^2; diameter: 25.4 mm). The laser was illuminated to the side-wall of a urea gel-filled container, and the back-scattered Raman photons were gathered by the detection probe (refer to Fig. 1).

In the transmission and APO measurements, a separate laser (Invictus, Kaiser, wavelength: 785 nm) was used to illuminate the same spot on the left wall of the container. The diameter of the laser illumination was approximately 1.0 mm. The same detection probe was positioned opposite the illumination spot and beneath the container bottom in the transmission and APO measurements, respectively. The axial offset distances were 13 mm. The diameter of the detection window in the WW-APO and NW-APO schemes was 25.4 and 10.0 mm, respectively. In the NW-APO scheme, an aluminum disk with a 10.0 mm hole was positioned in front of the detection probe (refer to Fig. 1).

The back-scattering spectra of samples were collected by accumulating 10 scans with 5 s exposure at each scan. Meanwhile, since the location of the detection probe was farther from the point of the laser illumination in the transmission and APO schemes, the number of generated Raman photons reaching the detector would decrease. Therefore, the exposure time was lengthened to 10 s in these measurements to increase the intensities of the Raman peaks to some degree. Even with the doubled exposure time, the overall peak intensities were approximately 16–25% of that in the back-scattering measurement. For all measurements, the laser power was maintained at 400 mW. For each sample, five replicate spectra were separately collected by randomly illuminating the laser at different locations on the left wall of the container. Baseline correction, normalization of spectra, and PLS were performed using MATLAB version 2017a (The Math Works Inc., MA, USA).

Monte Carlo simulation

As previously performed, the structure factor[4,16–19] was utilized for the Monte Carlo simulation. A more detailed description of the simulation can be found in the above reference, as well as our previous publications.[3,4,20–22] Table 1 shows the absorption and scattering coefficients of the urea gel and PP container used for the simulation. The number of illuminated laser photons was 100 000 in each measurement. The sizes of the laser illumination, detection window, and container in the simulation were the same as those in the experimental setup. The simulation was performed using a fully licensed Monte Carlo simulator (Version 2016).

Results and discussion

Raman spectra of samples acquired with back-scattering, transmission, and APO schemes

Fig. 2(a) shows the Raman spectra (1550–750 cm^{-1} range) of urea, propylene glycol, carbopol 934, and parabens (mixture of methyl 4-hydroxybenzoate and propyl 4-hydroxybenzoate). The strong urea peak at 1002 cm^{-1} corresponds to the N–C–N symmetric stretching mode,[23–25] and diverse peaks of propylene glycol and parabens appear over the spectral range. The peak of carbopol 934 is broad with low intensity (refer to the right *y*-axis scale). Although the small peaks of propylene glycol and parabens are near the 1002 cm^{-1} urea peak, they would not seriously interfere with the urea peak due to the low intensity. In addition, the concentrations of propylene glycol and parabens were constant in the urea gel samples. Although water is the most dominant component (over 78%) in the urea gel, its peak on a final sample spectrum is minute because water is a very weak Raman scatter.

Fig. 2(b) shows the back-scattering Raman spectra of the empty plastic container (black), 12.5 wt% urea gel in a quartz cuvette (red), and the same sample housed in the PP container

Table 1 Parameters of the PP container and urea gel used for the Monte Carlo simulation

	Absorption coefficient (cm^{-1})	Scattering coefficient (cm^{-1})	Refractive index
PP	0.0	0.07×10^{-3}	1.48
Urea gel	4.0	0.09×10^{-3}	1.49

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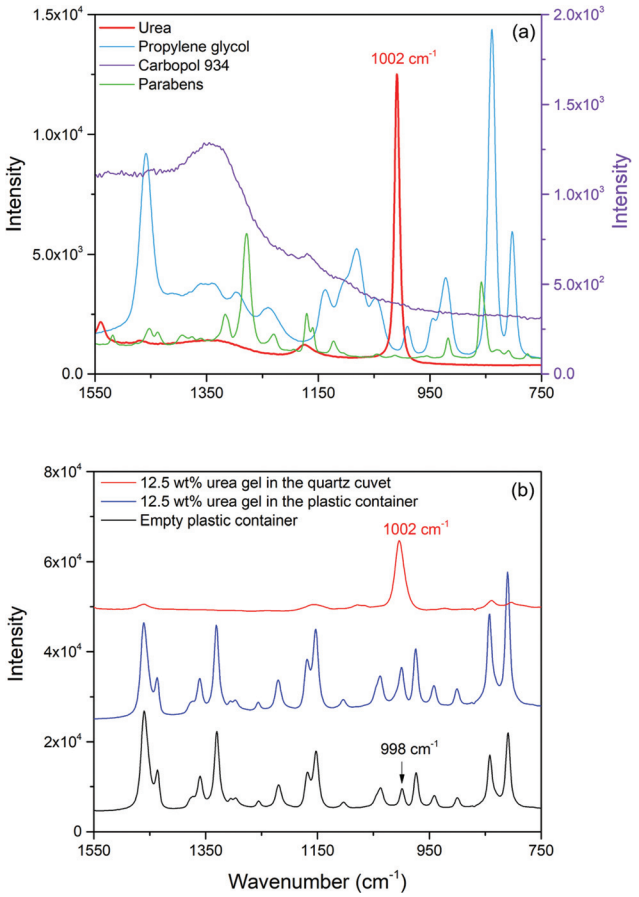

Fig. 2 Raman spectra of urea, propylene glycol, carbopol 934, and parabens (a). Back-scattering Raman spectra of 12.5 wt% urea gel in a 1 cm thick quartz cuvette, 12.5 wt% urea gel in the plastic container, and an empty plastic container (b).

(blue). The spectra are arbitrarily offset for a clear comparison. The urea peak (1002 cm^{-1}) is apparent when the sample is housed in the quartz cuvette. The empty plastic container shows numerous intense peaks over the spectral range, and the 998 cm^{-1} peak is nearest to the 1002 cm^{-1} urea peak. Therefore, overlap of the urea peak with the 998 cm^{-1} container peak is unavoidable and will hinder recognition of the urea peak. As expected, the spectrum of the urea gel in the container is quite similar to that of the empty container. Thus, a spectral acquisition scheme that is able to distinguish more urea information, while simultaneously reducing the container information, is necessary to secure an accurate determination of the urea concentration.

Since the spectrum of the urea gel-housed container in Fig. 2(b) was collected with the back-scattering mode, the container peaks are predominant, as expected. In this context, the transmission or APO scheme would be a better choice for the reduction of the interfering container peak. Since the locations of photon detection in these schemes are physically distant from the point of laser illumination, fewer Raman photons from the container that are generated at the wall will arrive at a detector. The linear distance between the laser illumination and photon detection in the transmission and APO schemes was 31.8 and 20.5 mm, respectively. Since both schemes unavoidably decreased the number of urea photons reaching the detector, the overall intensities of the peaks in the collected spectra were lower compared to that in the back-scattering measurement, as explained in the Experimental section.

Fig. 3(a), (b) and (c) shows the normalized back-scattering, transmission, and WW-APO spectra of all the samples in the 1190–750 cm^{-1} range, highlighting the urea peak at 1002 cm^{-1} and the strong container peak at 806 cm^{-1}. The spectra in the 1040–960 cm^{-1} range, where the urea band is centered, are magnified in the inset of each plot. Unit vector normalization was employed. Initially, the baseline of each spectrum was linearly corrected to zero at 6 different wavenumbers (1550, 1080, 1022, 987, 920 and 750 cm^{-1}). Then, the sum of the squares of all y-values (peak intensities) was calculated, and the baseline-corrected spectrum was divided by the square root of this sum for the normalization. The size of the detection window in these three measurements was 25.4 mm. In all cases, the container peaks are dominant, and the average intensity of the 806 cm^{-1} container peak in the back-scattering, transmission, and WW-APO spectra was 0.058, 0.055, and 0.055, respectively. The intensity of the container peaks in the back-scattering measurement was highest, and the intensity variation around 1002 cm^{-1} from the change of the urea concentration was smallest. The back-scattering spectra are more representative of the container wall, where the laser is illuminated. Therefore, the variation of the urea concentration is less sensitively recognized.

Although the decreases of the 806 cm^{-1} container peak in the transmission and WW-APO measurements were not significant (from 0.058 to 0.055), the intensity variations around the urea peak by the change of the urea concentration become substantially greater. Due to the distant positioning of the detector from a point of laser illumination in these measurements, the corresponding Raman sampling volume for the urea gel was greater, resulting in a larger intensity variation of the urea peak. Meanwhile, many container Raman photons still arrive at the detector, as seen in the collected spectra. For quantitative comparison, the intensity ratios of the 806 cm^{-1} container and 1002 cm^{-1} urea peak (designated as $I_{urea}/I_{container}$) in the three measurements were calculated using the spectra of the 7.5, 10.0, and 12.5 wt% samples. The results are shown in Table 2. For each sample, $I_{urea}/I_{container}$ was smallest in the back-scattering measurement, as expected. Meanwhile, $I_{urea}/I_{container}$ in the transmission measurement was greater than that in the WW-APO measurement. This is because the sampling volume in the transmission measurement was greater due to the longer distance between the laser illumination and photon detection.

Determination of urea gel concentration in each measurement and comparison of accuracies

PLS was used to determine the urea concentrations of the samples in each measurement within the 1190–750 cm^{-1} range with leave-one-out cross-validation. Fig. 4(a), (b) and (c) shows the concentration correlation plots (reference concen-

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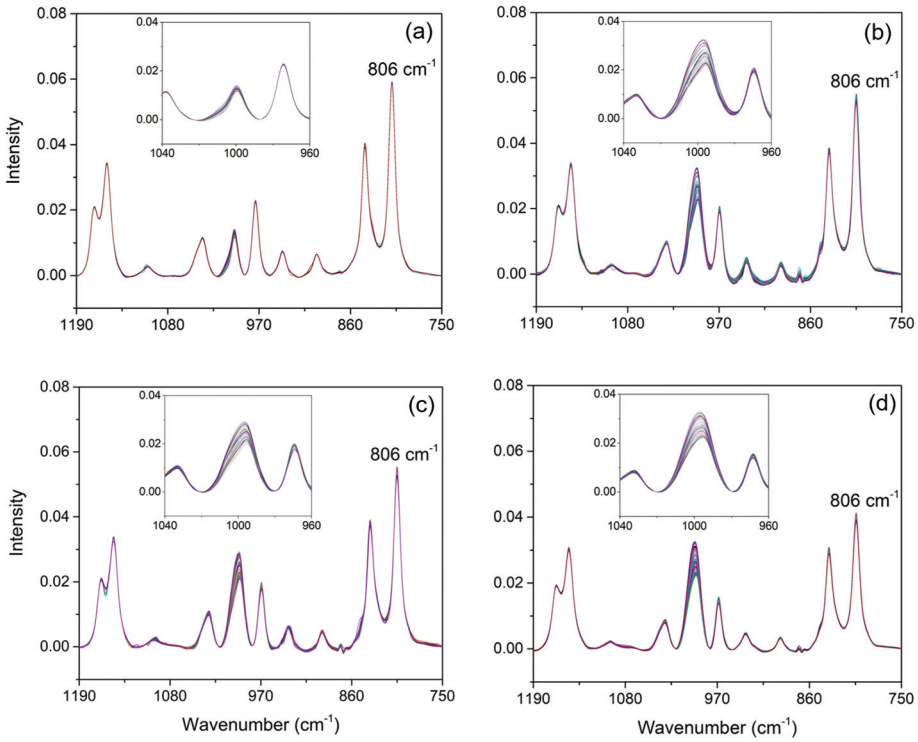

Fig. 3 Normalized Raman spectra of all samples measured with back-scattering (a), transmission (b), WW-APO (c), and NW-APO (d) schemes. The inset in each plot highlights the 1040–960 cm^{-1} range, containing the main urea band at 1002 cm^{-1}.

Table 2 $I_{urea}/I_{container}$ values calculated from the spectra of three urea gel samples acquired with the back-scattering, transmission, WW-APO, and NW-APO schemes

Employed scheme	7.5 wt% urea gel	10.0 wt% urea gel	12.5 wt% urea gel
Back-scattering	0.22	0.23	0.24
Transmission	0.40	0.47	0.58
WW-APO	0.38	0.46	0.55
NW-APO	0.56	0.66	0.82

tration *vs.* determined concentration by PLS) of the back-scattering, transmission, and WW-APO measurements, respectively. In all cases, two factors (latent vectors) were used in the PLS models. The corresponding standard errors of the cross-validation (SECV) and standard error of prediction (SEP) methods are designated in each plot. A notable observation is that the accuracy in the back-scattering measurement was substantially worse (SEP: of 0.75 wt%), and the determined concentrations of each sample (5 replicate measurements) were largely scattered. Since the spectral variation caused by the urea concentration change was small (Fig. 3(a)), the overlapped container peak complicated the recognition of the urea peaks in PLS modeling, degrading the accuracy.

In contrast, the SEP decreased to 0.44 and 0.35 wt% in the transmission and WW-APO measurements, respectively, due to the increased variation of the urea peak (Fig. 3(b) and (c)). Interestingly, although $I_{urea}/I_{container}$ was greater in the transmission measurement (Table 2), the SEP was slightly higher than that in the WW-APO measurement. Since the average travel distance of the urea photons to a detector was longer in the transmission measurement, the uncertainty of the photon propagation in the medium of the urea gel also increased. Due to this possibility, the concentration representation of the housed urea gel in the transmission measurement may be slightly inaccurate, explaining the observed inferior accuracy. An optimal Raman sampling volume leading to accurate analysis exists for a given sample system. Therefore, in this context, the WW-APO measurement seems to be optimal for the direct analysis of the housed urea gel.

One potential way to improve the accuracy would be to further decrease the container peak by rejecting more container Raman photons in the WW-APO measurement. For this purpose, the size of the detection window was decreased from 25.4 to 10.0 mm in the APO scheme (NW-APO), and the same samples were measured again. Fig. 3(d) shows all sample spectra collected using the NW-APO scheme. As seen, the intensity of the container peak (806 cm^{-1}) decreased to 0.041 (0.055 in the WW-APO measurement), since more of the container Raman photons generated at the side container wall were blocked from detection by narrowing the detection window. As a result, the corresponding $I_{urea}/I_{container}$ was greatest among the four measurements (Table 2), and the SEP further decreased to 0.31 wt% (Fig. 4(d)).

To investigate which spectral information was utilized in PLS modeling to determine the urea concentration in each

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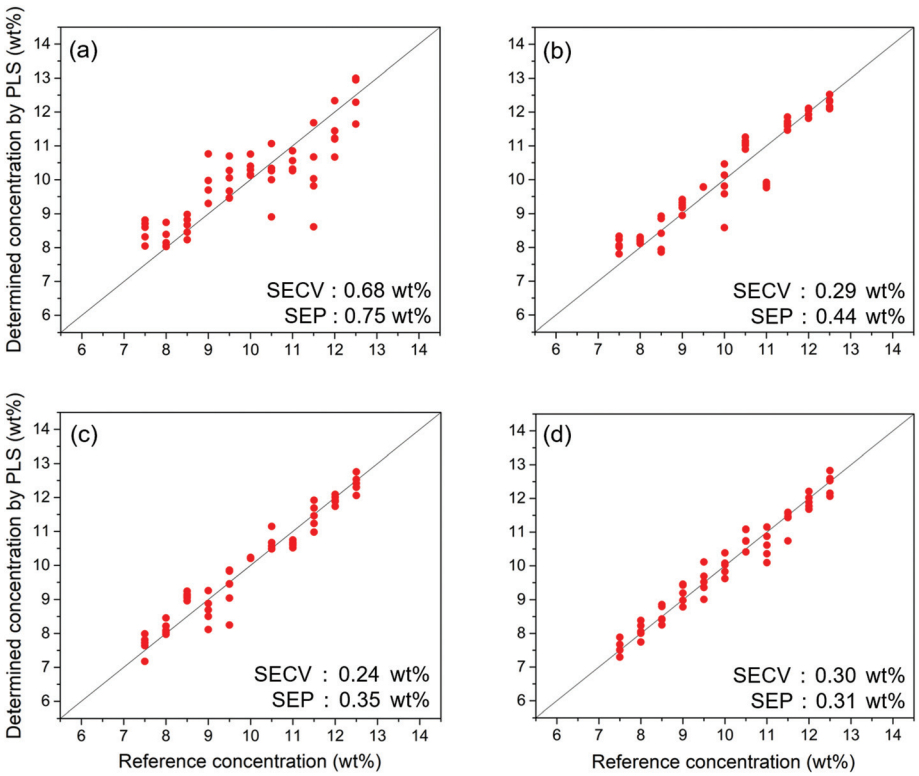

Fig. 4 Concentration correlation plots resulting from the determination of the urea concentration in back-scattering (a), transmission (b), WW-APO (c), and NW-APO (d) measurements.

case, the corresponding latent vectors (loadings) were compared together. Fig. 5 shows the first (black) and second (red) loadings in the PLS model in each measurement. The loadings of each measurement are arbitrarily offset for clarity. The vertical line indicates the 1002 cm^{-1}, where the peak of pure urea is centered. The most imposing observation in all measurements is that the first loadings are predominantly weighted on the peak around 1002 cm^{-1}, where the urea peak is overlapped with the 998 cm^{-1} container peak. In contrast, the weights on the other peaks in this range are considerably small. This observation confirms that each PLS model clearly targets the localized urea information for the concentration determination.

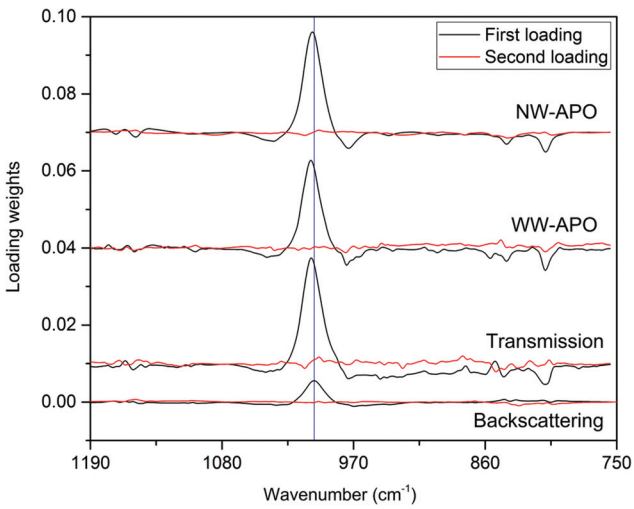

Fig. 5 First (black) and second (red) loadings in the PLS models employed to determine the urea concentration in the back-scattering, transmission, WW-APO, and NW-APO measurements. The loadings of each measurement are arbitrarily offset for clarity. The vertical line indicates the 1002 cm^{-1}, where the peak of pure urea is centered.

The magnitude of the loading weight is smallest in the back-scattering measurement since the variation of the urea peak according to the concentration change is smallest (refer to Fig. 3(a)). Meanwhile, it significantly increases in the other three measurements due to the greater presence of urea information in the corresponding spectra. In addition, the positions of the maximal loading weight in these three measurements move to the left compared to that in the back-scattering measurement. Since the urea peak is located at the left of the container peak, the shift of the loading weight to the left indicates the greater recognition of the urea information in the PLS modeling by the reduced container peak information, equivalent to the more accurate determination of the urea concentration. Nonetheless, the container information-free modeling is unrealizable as shown, and the co-appearing container peak influences the accuracy of the urea concentration determination.

To further investigate the origin of the improved accuracy in the NW-APO measurement, the standard deviations of 5 replicate predicted concentration values of each sample (a

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total of 11 samples) were calculated for both WW-APO and NW-APO measurements. The average standard deviation in the WW-APO and NW-APO measurements was 0.23 and 0.26 wt%, respectively, implying similar reproducibilities of the two measurements. Meanwhile, when SEP was calculated using an average predicted value of each sample instead of the 5 replicate values, it was 0.22 and 0.10 wt% in the WW-APO and NW-APO measurement, respectively. This finding shows that the accuracy enhancement achieved by using the NW-APO measurement is from the more correct Raman representation of the urea concentration through the alleviation of the interference of the container peak.

Monte Carlo simulation

For supplementary examination, a Monte Carlo simulation was performed to visualize the distributions of the container and urea Raman photons inside the container. The shape of the container was assumed to be cylindrical for easy simulation. Other than the shape, the physical dimensions of diameter, height, and wall thickness in the simulation were the same as those of the real container. Fig. 6(a) and (b) shows the distribution of the generated container and urea Raman photons, respectively, inside the container. The container photons were populated near the laser illumination spot and broadly spread inside the container. The considerable diffusion of the generated container photons inside the container wall due to scattering was responsible for the broad distribution throughout the container. In addition, the urea photons were relatively more populated at the left, while their distribution inside the container was not as broad as that of the container photons since the urea gel is somewhat transparent.

Fig. 6(c) and (d) shows the container and urea photons, respectively, distributed inside the 1 cm diameter cylinder in the middle, simulating the NW-APO measurement. In both cases, the photons were more localized at the left of the cylinder, while the container photons were more widely spread. Based on the simulation, the number of detected container photons in the NW-APO measurement decreased by 36.6% compared with that in the WW-APO measurement. The decrease in the urea photons was 32.1%. The narrowing detection window decreased the numbers of both container and urea photons arriving at the detector, while the decrease of the

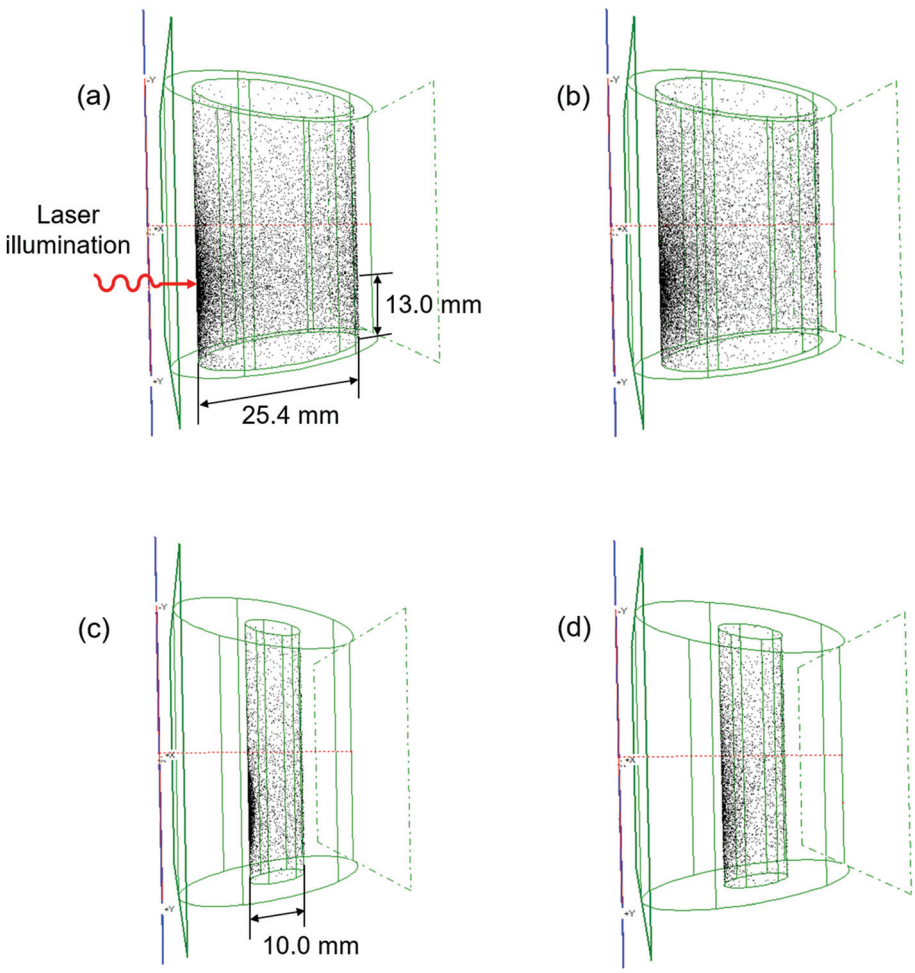

Fig. 6 Distribution of the generated container (a) and urea Raman photons (b) throughout the whole container. Distribution of the container (c) and urea Raman photons (d) in the middle of the 1 cm diameter cylinder.

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container photons was relatively more significant. In the actual spectra shown in Fig. 3(c) and (d), the intensity of the container peak (806 cm^{-1}) in the NW-APO measurement decreased by 25.4% compared with that in the WW-APO measurement (from 0.055 to 0.041). Conversely, the intensity of urea peak (1002 cm^{-1}) did not change significantly between both measurements. The simulation result partially agrees with the experimental observation.

Conclusion

In the direct and non-sampling Raman spectroscopic analysis of the housed sample in a plastic container, the suppression of the container information with minimal loss of sample features in a collected spectrum was key to securing accurate determination of the sample concentration. The NW-APO scheme was beneficial to reduce the intensity of the interfering container peak by narrowing the detection window in the APO arrangement, and ultimately enhanced the accuracy of the urea concentration determination. Meanwhile, the variation of the wall thickness among the employed containers could be a concerning issue since it directly influences the accuracy of the urea concentration determination. Nonetheless, the thickness variation is expected to be negligible because the containers were mass-produced plastic containers by injection molding. Overall, this study shows further interesting issues in the APO-based direct analysis of a housed sample in a container, since the accuracy is expected to depend on diverse parameters such as the container material, container size, wall thickness, sample turbidity and physical properties. Therefore, deeper evaluations of the accuracy of APO measurements by systematically varying the abovementioned parameters are under way in our research group. The inter-relationship of accuracy with the evaluated APO experimental parameters is of great interest.

Conflicts of interest

There are no conflicts to declare.

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